



UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Applied Econometrics (Theory)
Course Code: STAT-409

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SECTION (II)

Q.2. Define the following: (20)

- | | |
|--|-------------------------------|
| (i) Assumptions of M.L. estimation method for S.L.R. model | (ii) Over identified equation |
| (iii) Pareto Curve | (iv) Auto correlation |
| (v) Distributed Lag model | (vi) Multicollinearity |
| (vii) Statement of Aitken Theorem | (viii) Stochastic Regression |
| (ix) Rank Condition of identification | (x) Orthogonal Polynomials |

Q.3. Discuss the consequences of using OLS estimators in the presence of first order auto-correlated error terms. (06)

Q.4. For the model (12)

$$y_{1t} = \beta_{12} y_{2t} + \gamma_{11} x_{1t} + u_{1t}$$

$$y_{2t} = \beta_{21} y_{1t} + \gamma_{22} x_{2t} + \gamma_{23} x_{3t} + u_{2t}$$

You are given the following information:

1. The least square estimates of the reduced-form coefficients are

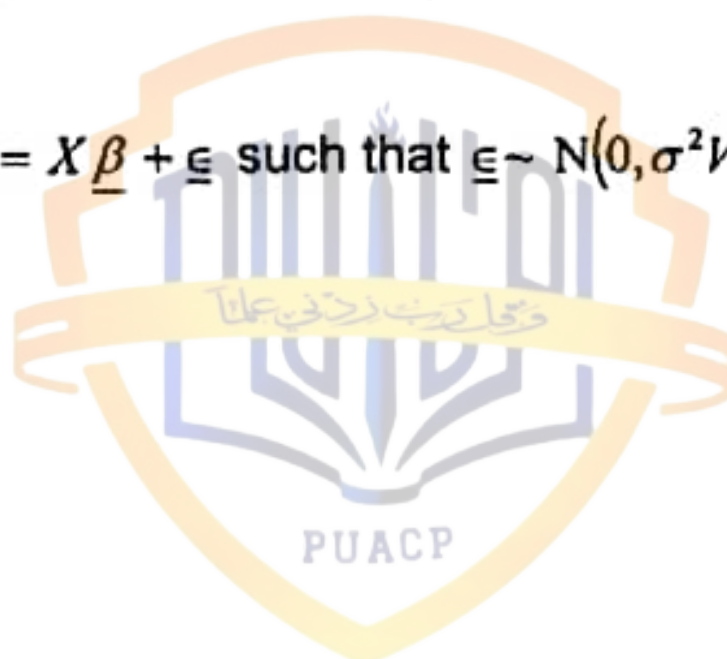
$$\begin{bmatrix} 5 & 10 & 2 \\ 10 & 10 & 5 \end{bmatrix}$$

2. The estimates of variances of the regression coefficients in the first reduced-form equation are 1, 0.5, 0.1.
3. The corresponding covariance's are estimated to be all zero.
4. The estimated variance of the error on the first reduced-form equation is 2.0.

Compute the consistent estimates of structural parameters of the first structural equation.

Q.5. Define Heteroskedasticity and discuss two measures to remove / minimize it. (06)

Q.6. Let $\underline{Y} = \underline{X}\underline{\beta} + \underline{\epsilon}$ such that $\underline{\epsilon} \sim N(0, \sigma^2 \underline{V})$ obtain BLUE of σ^2 . (06)





ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Discuss briefly the following:

- (i) Assumptions of generalized least squares
- (ii) Heteroskedasticity
- (iii) Instrumental variable
- (iv) Indirect Least squares method
- (v) Geary test for Autocorrelation

(20)

Q.3. Following are the OLS residuals obtained by fitting the model
 $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$
19.98, 11.50, -1.78, 0.33, -6.05, -14.83, -14.12, -14.39, -9.17, -0.36, 4.86, 1.68, 9.10, 10.32, 2.93

Test autocorrelation by Geary test and estimate first order autocorrelation Co-efficient by Cochran-ortcutt two step procedure.

(10)

Q.4. Consider the model:

$$y_1 = \beta y_2 + u_1, \quad y_2 = \alpha y_1 + \alpha_2 x_1 + \alpha_3 x_2 + u_2$$

Obtain consistent estimates of structural parameters $(\beta, \alpha_1, \alpha_2, \alpha_3)$, where possible, by appropriate method using the calculations.

$$\Sigma x_1^2 = 1, \quad \Sigma x_2^2 = 20, \quad \Sigma x_1 x_2 = 0, \quad \Sigma x_1 y_1 = 5, \quad \Sigma x_2 y_1 = 40,$$

$$\Sigma x_1 y_2 = 10, \quad \Sigma x_2 y_2 = 20$$

(10)

Q.5. For GLR model $Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + u$ such that $\text{Var}(u_i) \propto X_{3i}$, residual SS for first 30 observations is 377.17 and residual SS for last 32 observations is 1536.80. Test the Heteroskedasticity.

(05)

Q.6. Discuss the practical consequences of Multicollinearity.

(05)





UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program / Eighth Semester – 2020

Roll No. _____

Paper: Applied Econometrics (Theory)
Course Code: STAT-409 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

(4x5=20)

Q.2. Answer the following short questions.

- i. Dummy Variables.
- ii. ²⁴⁵ Simultaneous equation system.
- iii. Orthogonal Polynomials and their use. ¹⁶⁰
- iv. Reasons of Autocorrelation.

Answer the following questions.

(3x10=30)

- i. ²³² What is instrumental variable? Show that the instrumental variables estimators are consistent.
- ii. State and prove Aitken theorem. ¹⁴²
- iii. Define Heteroskedasticity and discuss some measures to remove it. ¹⁹⁰





UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Applied Econometrics (Theory)

Course Code: STAT-409

Roll No. 226473

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers to the following questions.

(6x5=30)

- a) Specification Error.
- b) Park test.
- c) Simultaneous Equation Bias.
- d) Difference between perfect and imperfect multicollinearity.
- e) Assumptions of MLE method.
- f) Difference between simultaneous and recursive equations model.

Write detailed answers to the following questions.

(3x10=30)

Q.2. Discuss the Gold-Feld Quandt test of heteroscedasticity.

Q.3. What corrective measures should be taken to avoid multicollinearity?

Q.4. Consider the following data and test autocorrelation by Durbin Watson. Test at 5% level of significance ($d_L = 0.697, d_U = 1.64$)

3.03567 > 0.697

Y	4	4	5	6	7	9	8	10	12	14
X ₁	15	18	25	40	50	55	60	70	80	90
X ₂	16	12	10	8	8	6	5	4	3	2
error	0.16	0.43	0.12	-0.22	-0.45	1.25	-1.3	-0.24	-0.45	1.09





UNIVERSITY OF THE PUNJAB

B.S. 4 Year Program / English Medium

Paper: Applied Econometrics (Theory)

Course Code: STAT-409

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6×5=30)

- Explain assumptions of Generalized Least Square.
- Write steps of Gold Fied Quandt Watson Test.
- Write about order and rank conditions for identification.
- Write the names of tests for the detection Heteroscedasticity.
- Write the reasons for autocorrelation.
- Write about the Indirect Least Square Method.

Answer the following questions.

(3×10=30)

Q.2. Discuss the detection and consequences of Multicollinearity.

Q.3. Consider the structural equations model : $Y_1 = \alpha_1 Y_2 + \alpha_2 X_1 + u_1$ and $Y_2 = \beta_2 Y_1 + \beta_3 X_2 + u_2$ produces the following reduced form equations: $Y_1 = 1.5X_1 + 2.5X_2 + V_1$ and $Y_2 = 2.5X_1 + 3.5X_2 + V_2$.

Determine the structural parameters from the estimated reduce form coefficient.

Q.4. Following are the OLS residuals obtained by fitting the model.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$$

19.96, 11.50, -1.78, 0.33, -6.05, -14.83, -14.12, -14.39, -9.17, -0.36, 4.86, 1.68, 9.10, 10.32, 2.93.

Test autocorrelation by Geary test and estimate first order autocorrelation coefficient by Cochran- Orcutt two step procedure.